Geometry Review: Unit 1 — Basics & Reasoning

Answer Key

Station 1: Segments, Midpoints, and Distance

1. $AC = 19$
2. $BC = 16$
3. $x = 7,\ AB = 15,\ BC = 17$ $\quad 5x - 3 = 32$
4. $x = 6,\ AB = 54$ $\quad 5x - 3 = 3x + 9;\ AM = 27$
5. $(5, 1)$
6. $B(5, 7)$
7. 10 $\quad \sqrt{6^2 + 8^2}$
8. $3\sqrt{5} \approx 6.7$ $\quad \sqrt{36 + 9} = \sqrt{45}$
9. $PQ = 12;$ midpoint $= 3$
10. $XZ = 12,\ ZY = 18$

Station 2: Angle Relationships and Angle Bisectors

1. acute, right, obtuse, straight
2. $28°$
3. $65°$
4. $85°$
5. $38°$
6. $25°,\ 65°$ $\quad 3x + 15 = 90$
7. $86°,\ 94°$ $\quad 10x - 10 = 180,\ x = 19$
8. $60°$ $\quad 4x - 6 = 2x + 12,\ x = 9$
9. $90°$
10. $30°$ $\quad 180 - x = 2(90 - x) + 30$

Station 3: Conditional Statements and Their Related Forms

1. Hypothesis: it rains. Conclusion: the ground is wet.
2. If a number is even, then it is divisible by 4.
3. False; counterexample: 6 is even but not divisible by 4
4. If $x \leq 5$, then $x \leq 3$.
5. If $x \leq 3$, then $x \leq 5$.
6. Both are true (a conditional and its contrapositive always have the same truth value)
7. 2 is prime but even
8. If a figure is a square, then it has four sides.
9. An angle measures $90°$ if and only if it is a right angle.
10. If a triangle is not isosceles, then it is not equilateral.

Station 4: Inductive and Deductive Reasoning

1. $48,\ 96$ $\quad$ *multiply by 2*
2. $14,\ 17$ $\quad$ *add 3*
3. $36,\ 49$ $\quad$ *perfect squares*
4. $1,\ \dfrac{1}{3}$ $\quad$ *divide by 3*
5. $3n + 1;\ 151$
6. Inductive; deductive
7. Mia will pass.
8. If it snows, then we sleep in.
9. $5 + (-2) = 3$, which is not greater than 5
10. $15;\ \dfrac{n(n-1)}{2}$